

Insight, part of a Special Feature on [Beyond the Assessment on the Diverse Values of Nature: Hidden gems, Biases, Frontiers, Challenges, and Insights](#)

Making room for meaningful inclusion of Indigenous and local knowledge in global assessments: our experiences in the values assessment of the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services

Mariaelena Huambachano¹ , Gabriel R. Nemogá Soto^{2,3}  and Tuyeni H. Mwampamba^{4,5,6} 

ABSTRACT. In recognizing the urgent need to address global challenges such as biodiversity loss, food insecurity, and climate change, it is essential to incorporate diverse knowledge systems, including Indigenous worldviews and knowledge of nature. The Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES) has taken significant steps to integrate Indigenous and local knowledge (ILK) as well as the viewpoints of Indigenous peoples and local communities (IPLC) into its thematic and methodological assessments. This inclusive approach enriches our understanding of nature and enhances our ability to tackle these pressing global issues. The recently published IPBES Report on Diverse Values and Valuation of Nature, also known as the values assessment (VA), includes Indigenous scholars and ILK experts as authors. The VA provides an interdisciplinary synthesis of existing knowledge on the various ways in which humans value nature, as well as the methods and approaches for understanding these values. It also examines the extent to which these values are integrated into decision-making structures and processes. We are a group of Indigenous scholars and ILK experts from the Global South who participated in the VA, specifically in Chapter 3's "ILK Team." The value of including IPLC in knowledge-synthesis initiatives is highlighted by our experiences. There are opportunities to improve the inclusion of ILK in similar assessments. The lessons we learned while working at the VA have motivated us to recommend that future assessments and similar initiatives should actively involve IPLC, their knowledge systems, and their ancestral wisdom.

Key Words: *indigenous and local knowledge; indigenous peoples and local communities; indigenous worldviews; intergovernmental science-policy platform on biodiversity and ecosystem services; values and valuation assessment; western science*

INTRODUCTION

In 2012 the plenary of the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES) was established as an independent intergovernmental body composed of 94 member governments at the time, which has increased to almost 150 members today (<https://www.ipbes.net/about>). Since its establishment, IPBES has committed to "recognize and respect the contribution of Indigenous and local knowledge to the conservation and sustainable use of biodiversity and ecosystems" as one of its 11 operating principles (Hill et al. 2020:9). In 2017 IPBES member states concretized this commitment by adopting the IPBES ILK Approach, which provides a framework for engaging Indigenous peoples and local communities (IPLC) and Indigenous local knowledge (ILK) in all four functions of IPBES, including the scoping, production, and review of assessments (IPBES 2017). The framework recognizes the need for assessments to engage with three types of experts: ILK holders, ILK experts, and experts on ILK (IPBES 2017). IPBES's recent Assessment of the Diverse Values and Valuation of Nature (also known as the Values Assessment and hereafter the VA) explores how people conceptualize human-nature relationships across different regions and social contexts. Supporting IPBES's efforts to include ILK in assessments, the VA was committed to incorporating scientific and ILK knowledge. The VA was exceptional in recruiting a diverse set of authors to undertake the

assessment. Authors from various social-science backgrounds, Global South epistemologies, and Indigenous scholarship, in addition to natural scientists, were represented in the group. We, a team of Indigenous and ILK experts from Peru, Colombia, and Tanzania, served as lead authors for Chapter 3 of the VA from 2019 to 2022. Our task in Chapter 3 was to evaluate valuation and valuation methods, with a focus on incorporating IPLC knowledge, synthesizing information and knowledge about IPLC valuation approaches and practices.

Literature on ILK and traditional ecological knowledge is often written by non-Indigenous scientists and there is a long history of distortion, misrepresentation, and manipulation of ILK "to fit within the desired [i.e., Western science] frameworks, definitions, and conceptualizations" (Simpson 2004:376). IPLC valuation is an uncharted territory in valuation and Indigenous studies. We observed the significant influence of Western science on defining valuation concepts and the desire among most authors for IPLC ways to fit neatly into pre-existing frameworks. When applied to IPLC contexts, Western science valuation concepts and terminology such as "specific and broad values," "plural valuation," and "value dimensions" were unfamiliar and awkward to use for the 26 contributing authors who supported Chapter 3 ILK. As part of the deeply rooted obligations we have toward our respective communities, we were compelled to make

¹Syracuse University, Syracuse, New York, USA, ²University of Winnipeg, Winnipeg, Manitoba, Canada, ³Research Group on Biodiversity, Genetic Resources, and Traditional Knowledge (PLEBIO), National University of Colombia, Bogotá, Colombia, ⁴The Nature Conservancy, Africa Program, Arusha Office, Arusha, Tanzania, ⁵Department of Ecosystems and Conservation, College of Forestry, Wildlife, and Tourism, Sokoine University of Agriculture, Morogoro, Tanzania, ⁶Institute for Ecosystems and Sustainability Research, National Autonomous University of Mexico, Morelia Campus, Morelia, Michoacan, Mexico

sure that our work in IPBES did not replicate historical distortions of ILK. Therefore, we saw our involvement in IPBES and the VA as a special chance to push for fair inclusion and representation of ILK, and to establish balanced relationships between IPLC and Western scientific knowledge systems.

We found that the methods and practices developed in academia did not directly apply to IPLC contexts (Termansen et al. 2022). Tense conversations with our fellow authors were unavoidable. However, disagreements led us to explore the sources of the tension and, from that, better articulate the opportunities and limits of integrating Indigenous valuation systems. In this paper, we reflect on critical lessons from this experience, which can inform similar science-policy efforts committed to broader engagement of IPLC, particularly in informing global biodiversity conservation and ecosystem management. We summarize these lessons into five challenges and four revelations, or hidden gems, that best capture our learning. Our insights complement other authors' reflections on IPBES's processes to include ILK by focusing on our experiences as ILK and IPLC authors embedded in global assessments (see, e.g., Hill et al. 2020, McElwee et al. 2020).

Challenge 1: Structural limitations inhibit meaningful participation of IPLC

The intentions of IPBES to incorporate the knowledge of IPLC in its assessments are commendable. However, its expert recruitment policies have limitations. Interested experts must go through their government's focal point or an approved organization to contribute. Authors are primarily selected based on academic merit, which puts ILK holders who are not proficient in English at a disadvantage. Consequently, perspectives from the Global South are largely missing from the assessments. The IPBES ILK Taskforce organizes ILK dialogues to integrate IPLCs' perspectives, but we found these dialogues to be limited in scope.

Challenge 2: The minority tax borne by Indigenous scholars and ILK experts

The term "minority tax" refers to the additional burden faced by individuals from underrepresented groups in workplace settings, especially academia (Faucett et al. 2022). This includes extra responsibilities such as serving on diversity committees, mentoring other minority employees and students, and providing perspectives on racial or cultural issues (Kamceva et al. 2022). These responsibilities come with emotional, psychological, and time-based costs, diverting energy from primary job responsibilities (Campbell and Rodríguez 2019, Akin 2020, Kamceva et al. 2022, Mwampamba et al. 2022).

We needed to prepare for the time and energy to justify our positionality as Indigenous scholars and ILK experts and negotiate an alternative model for working with ILK in Chapter 3. Numerous meetings were held among co-authors, sometimes with co-chair intervention, to discuss how to streamline ILK. After many of those meetings, we had additional meetings to strategize and encourage each other to keep going. The threat of quitting the VA process lurked in our minds, but the fear of giving up on such an essential process for IPLC inclusion rallied us forward. Despite a shared desire with other chapter authors to

work seamlessly with all knowledge systems, we found that forming an ILK Team within the chapter was the most productive way to move forward, even if it meant some isolation.

By framing these experiences as a minority tax, we seek to draw attention to a problem well-documented in the literature and experienced in numerous professional settings. Recognizing that these differences disproportionately affect underrepresented experts, we must strive to create better working environments for everyone. The IPBES process assumes that co-chairs, coordinating lead authors, and lead authors can adeptly manage these complex collaborations that bridge disciplines, cultures, and personalities. We firmly believe that incorporating ILK awareness and education into the IPBES onboarding process for all assessment authors and providing additional support to underrepresented groups can address this challenge.

Challenge 3: Conflating IPLC and over-generalizing place-based knowledge

IPBES's glossary defines IPLC as "ethnic groups who are descended from and identify with the original inhabitants of a given region, in contrast to groups that have settled, occupied, or colonized the area more recently" (https://files.ipbes.net/ipbes-web-prod-public-files/inline-files/IPBES_ILK_MethGuide_MEP-Approved_5MAY2022.pdf). This definition is problematic for Afro-descendant people, for example, who were forcibly transferred from Africa to the Americas and have inhabited new territories for centuries. Moreover, conflating Indigenous peoples with local community knowledge risks bypassing distinct Indigenous rights, such as the right to self-determination and to their cultural heritage and intellectual property, as outlined in articles 3 and 31, among others, of the United Nations Declaration on the Rights of Indigenous Peoples (UNDRIP). UNDRIP sets the first universal and transnational standards for Indigenous survival, dignity, well-being, and political and cultural rights.

During development of the VA, requests to decouple Indigenous peoples from local communities when assessing ILK literature were not well-received. Indigenous scholar Gabriel Nemogá (2019) proposed the definition of Indigenous Peoples and Like-Minded Local Communities (IPLMLC) to ensure that communities with long histories in the landscape who are oriented by short-term gains and colonizing enterprises on Indigenous lands were not counted as IPLC. Our assessment of IPLC valuation methods, approaches, and practices revealed that communities with worldviews similar to those of Indigenous people, i.e., like-minded, tended to have similar approaches to valuation.

IPBES 2022 recognizes the complexities of working with fixed definitions of Indigenous peoples and Indigenous knowledge as well as of local communities and local knowledge. The call is to avoid overgeneralizations in regions or territories, even when referring only to Indigenous peoples and Indigenous knowledge, and to instead account for the context-specific nature of IPLCs' knowledge systems. Overgeneralizations extend to the often-repeated idea that Indigenous and local communities live harmoniously with nature: the disruption of IPLCs' ways of life does not always allow for preserving sustainable uses and respectful practices with the non-human world.

Challenge 4: Limited capacity to process and synthesize IPLC and non-written knowledge with IPLC epistemologies

With very little published literature on IPLC valuation, we solicited essays on valuation from 26 ILK experts and undertook content analysis. The VA's Technical Support Unit (TSU) were instrumental for processing and analyzing the data, which generated key initial insights into IPLC valuation. However, the underlying epistemologies informing these analyses leaned heavily on Anglo-Eurocentric approaches for data analysis; there was eagerness to find patterns and processes that fit into existing categorizations and frameworks. Adopting solely this understanding of IPLC valuation would have undermined our responsibility to promote other ways of understanding the information shared by the ILK experts. Hence, we analyzed the same data using an Indigenous science approach, which uncovered bountiful insights that we did not have before. With this, we avoided a single-eyed western science approach and demonstrated the value of two-eyed seeing (Reid et al. 2020). Including IPLC authors in every assessment chapter is desirable but not always possible. The technical support unit traverses chapters, however, and is a key support to authors. This challenge highlights the need to recruit ILK experts within the TSU to help chapters to streamline working with ILK and support with knowledge co-generation.

Challenge 5: The location of ILK dialogues and formats for interactions limit meaningful outcomes

IPBES holds up to three ILK dialogues for each assessment per its ILK approach. The dialogues are workshops between assessment authors and IPLCs intended to facilitate the incorporation of ILK holders' perspectives on the different topics assessments address. Some of us attended these dialogues and reviewed 15 reports of dialogues of previous assessments; they are well-intentioned conversations that have become highly repetitive regarding the IPLC topics discussed and the information generated. The first VA ILK dialogue was held in March 2019 in Paris, the second was held in September 2019 in Oaxaca, an Indigenous community in Mexico, and the third, under COVID-19 travel restrictions, was held online in 2020. These dialogues were an acknowledgement of the limitations imposed by scarcity of ILK literature written from IPLC perspectives, and the fact that most ILK is not in written form and not in English.

The initial VA ILK dialogue, which took place in Paris without any IPLC context, felt more like an extractive conversation than a dialogue. It seemed to be designed to gather responses to pre-determined questions in order to fulfill a scientific inquiry influenced by perspectives from the Global North (IPBES 2019). The third online meeting lasted two hours and focused on gathering feedback for improving the VA process. We were unsure if we effectively communicated valuation and were frustrated because it was our final opportunity to provide input before the assessment's final draft.

The ILK dialogues are well-intentioned, but they should happen earlier in the assessment process, right after the first author meeting so that they can inform assessment framings before these become concretized. Rather than a momentary injection of IPLC input, the ILK dialogues could be designed as two-way conversations with diverse contributing authors that occur each

time key ILK advancements are made in the assessment. This approach holds the potential to significantly improve the assessment process.

We view these five challenges as unique opportunities for us, VA authors, IPBES, and other global assessment organizers to learn and grow as scientists and individuals in collaborative processes. These challenges underscored hidden gems and additional opportunities for reflection and learning that can guide future assessments.

Hidden Gem 1: Moving toward co-valuation of nature, rather than integration of valuation methods

Initial efforts to fit Indigenous valuation practices and approaches into the valuation method families proposed in Chapter 3 (Termansen et al. 2022) risked misrepresenting important Indigenous specificities of valuation. For example, Western-science informed statement-based methods only take into account the statements of humans and mostly humans currently occupying the earth. Indigenous approaches, however, consider the statements, i.e., messages, of non-human and non-material agents, future and past generations along with current humans including children and the elderly. Consequently, some Indigenous methods are spiritual and ceremonial and serve multiple purposes simultaneously, such as communal gatherings, value expressions, affirmations of responsibility to nature, and decision-making processes. This revelation allowed us to establish a space where Indigenous and non-Indigenous worldviews could converge on equal footing and where co-valuation, i.e., sense co-production of knowledge (Tengö et al. 2014), is possible. If grounded in meaningful and respectful complementarity, "co-valuation of nature, rather than integration, promotes a suitable space for bringing together multiple value dimensions and worldviews" (Termansen et al. 2022:73; see also MacGregor 2004, Kimmerer 2015, Huambachano 2023, Pascual et al. 2023). As other Indigenous scholars have pointed out, Indigenous worldviews and ILK should not be treated just as a body of knowledge that offers a new frontier for discovering, filtering, and extracting valuable data.

Hidden Gem 2: In union there is strength

The co-chairs and coordinating LAs at IPBES were supportive and approachable during the peer review and assessment process. They encouraged an inclusive working culture and were committed to incorporating multiple knowledge systems into the assessment. In the VA, support was given to the ILK team to highlight the unique aspects of IPLCs' valuation of nature, including an alternative approach informed by Indigenous peoples' worldviews. This approach aimed to show the differences between Indigenous and Anglo-Eurocentric approaches, highlighting how the Western science approach can misrepresent IPLCs' ways of knowing. For instance, it can misinterpret talking circles, putting them into the deliberative valuation category, where a group of people express their values. In reality, talking circles within ILK are rooted in traditions and have specific rituals guiding who can speak, when, and how.

Hidden Gem 3: Time is not money; taking time for deep and meaningful conversations

IPBES work often feels rushed, resulting in a work culture that operates on the assumption that time is linear, limited, and cannot be wasted. Speaking fast, eloquently, and concisely is highly

valued in such a work culture. While in some way efficient, this is a very specific communication style that clashes with styles that value slower conversations, listening to the opinions of everyone, even if repetitive, and giving space for silence. To address this, the ILK team scheduled longer meetings to allow for in-depth discussions without feeling rushed. In one instance, we scheduled a Zoom call without a set end time to ensure we could cover the topic thoroughly, and the meeting lasted four hours. This approach of holding meetings with no set end time is common in IPLC settings. Unrushed conversations are crucial in resolving misunderstandings, making team members feel more confident and secure in their roles and relationships among team members and the assessment process, especially when managing people in assessments.

Hidden Gem 4: In-situ engagement promotes cultural inclusion and meaningful participation

The second VA ILK dialogue in Capulálpam de Méndez, Oaxaca, showed the value of including ILK worldviews and ancestral wisdom. Free, prior, and informed consent was obtained for the dialogue, leading to a more intimate and transparent dynamic. This dialogue involved Indigenous leaders and knowledge holders from four IPBES regions and 11 Indigenous nations. Participants spoke Spanish and English, enabling an atmosphere for accessible communication and trust. In this ILK dialogue, it was notable that participants shared stories and narratives to communicate practices about the interrelations between their communities and nature. People also shared communities' experiences in their struggles to gain recognition and respect from national governments for their Indigenous traditions, practices, and collective rights. The stories emphasized an Indigenous worldview that includes human and non-human beings, promoting respect, reciprocity, and care as fundamental principles for individuals and the community. The Indigenous caucus successfully brought together knowledge holders from various parts of the world, creating a welcoming atmosphere and encouraging active participation in ceremonies and appreciation of the Indigenous-preserved ecosystem. The results and satisfaction of the ILK dialogue in Oaxaca were fulfilling on both individual and collective levels, in contrast to the limitations of other ILK dialogues.

CONCLUSION

The main lesson of the VA was that IPBES assessments can succeed if we turn challenges into hidden gems. We learned that participating ILK experts and knowledge holders in IPBES assessments ensure their worldviews and knowledge are clearly understood, respected, and included in future assessments. Therefore, we advocate for including IPLC knowledge systems in such processes. After all, many decisions about nature are made daily in IPLC territories, drawing from ILK. Favorable conditions are needed to support these experts in operating at their best. In future assessments, the IPBES ILK approach should be conveyed to all authors during the author meetings by dedicating specific sessions to the all-author meeting programs. Allocating an ILK TSU team to all IPBES assessments could ease ILK experts' responsibility to educate their co-authors on their roles and duties and acquire, process, and analyze ILK materials. In-situ ILK dialogues promote trust, enlightening participation, and joy. We

hope our personal experiences and reflections can support IPBES' future assessments and build pathways to engage meaningfully with IPLC, their knowledge, and their ancestral wisdom.

LITERATURE CITED

- Akin, Y. 2020. The time tax put on scientists of colour. *Nature* 583(7816):479-481. <https://doi.org/10.1038/d41586-020-01920-6>
- Campbell, K. M., and J. E. Rodríguez. 2019. Addressing the minority tax: perspectives from two diversity leaders on building minority faculty success in academic medicine. *Academic Medicine* 94(12):1854-1857. <https://doi.org/10.1097/ACM.0000000000002839>
- Faucett, E. A., M. J. Brenner, D. M. Thompson, and V. A. Flanary. 2022. Tackling the minority tax: a roadmap to redistributing engagement in diversity, equity, and inclusion initiatives. *Otolaryngology – Head and Neck Surgery* 166(6):1174-1181. <https://doi.org/10.1177/01945998221091696>
- Hill, R., Ç. Adem, W. V. Alangui, Z. Molnár, Y. Aumeeruddy-Thomas, P. Bridgewater, M. Tengö, R. Thaman, C. Y. A. Yao, F. Berkes, and J. Carino. 2020. Working with Indigenous, local, and scientific knowledge in assessments of nature and nature's linkages with people. *Current Opinion in Environmental Sustainability* 43:8-20. <https://doi.org/10.1016/j.cosust.2019.12.006>
- Huambachano, M. 2023. Seeding a movement: Indigenous food sovereignty. *University of Miami Law Review* 78:390.
- Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES). 2017. Report of the plenary of the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services on the work of its fifth session. IPBES, Bonn, Germany.
- Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES). 2019. Report from the first Indigenous and local knowledge dialogue workshop for the IPBES assessment of diverse conceptualizations of multiple values of nature. IPBES, Bonn, Germany.
- Kamceva, M., B. Kyerematen, S. Spigner, S. Bunting, S. Li-Sauerwine, J. Yee, and M. Gisondi. 2022. More work, less reward: the minority tax on U.S. medical students. *Journal of Wellness* 4 (1):5. <https://doi.org/10.55504/2578-9333.1116>
- Kimmerer, R. 2015. Braiding sweetgrass: Indigenous wisdom, scientific knowledge, and the teachings of plants. Milkweed Editions, Minneapolis, Minnesota, USA.
- McElwee, P., Á. Fernández-Llamazares, Y. Aumeeruddy-Thomas, D. Babai, P. Bates, K. Galvin, M. Guèze, J. Liu, Z. Molnár, H. T. Ngo, and V. Reyes-García. 2020. Working with Indigenous and local knowledge (ILK) in large-scale ecological assessments: reviewing the experience of the IPBES Global Assessment. *Journal of Applied Ecology* 57(9):1666-1676. <https://doi.org/10.1111/1365-2664.13705>

McGregor, D., 2004. Coming full circle: Indigenous knowledge, environment, and our future. *American Indian Quarterly* 28 (3/4):385-410. <https://doi.org/10.1353/aiq.2004.0101>

Mwampamba, T. H., B. N. Egoh, I. Borokini, and K. Njabo. 2022. Challenges encountered when doing research back home: perspectives from African conservation scientists in the diaspora. *Conservation Science and Practice* 4(5):564. <https://doi.org/10.1111/csp2.564>

Nemogá, G. 2019. Indigenous agrobiodiversity. *Agrobiodiversity: integrating knowledge for a sustainable future* 24:241. <https://doi.org/10.7551/mitpress/11989.003.0018>

Pascual, U., P. Balvanera, C. B. Anderson, R. Chaplin-Kramer, M. Christie, D. González-Jiménez, A. Martin, C. M. Raymond, M. Termansen, A. Vatn, and S. Athayde. 2023. Diverse values of nature for sustainability. *Nature* 620(7975):813-823. <https://doi.org/10.1038/s41586-023-06406-9>

Reid, A. J., L. E. Eckert, J. F. Lane, N. Young, S. G. Hinch, C. T. Darimont, S. J. Cooke, N. C. Ban, and A. Marshall. 2021. “Two-Eyed Seeing”: an Indigenous framework to transform fisheries research and management. *Fish and Fisheries* 22(2):243-261. <https://doi.org/10.1111/faf.12516>

Simpson, L. R., 2004. Anticolonial strategies for the recovery and maintenance of Indigenous knowledge. *American Indian Quarterly* 373:384. <https://doi.org/10.1353/aiq.2004.0107>

Tengö, M., E. S. Brondizio, T. Elmqvist, P. Malmer, and M. Spierenburg. 2014. Connecting diverse knowledge systems for enhanced ecosystem governance: the multiple evidence base approach. *Ambio* 43:579-591. <https://doi.org/10.1007/s13280-014-0501-3>

Termansen, M., S. S. Jacobs. T. H. Mwampamba. S. Ahn. S. A. Castro. N. Dendoncker, N. H. Ghazi, M. Gundimeda, M. Huambachano, H. Lee, et al. 2022. Chapter 3: The potential of valuation. Pages 123-245 in P. Balvanera, U. Pascual, M. Christie, B. Baptiste, and D. González-Jiménez, editors. *Methodological assessment report on the diverse values and valuation of nature of the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services*. IPBES, Bonn, Germany.